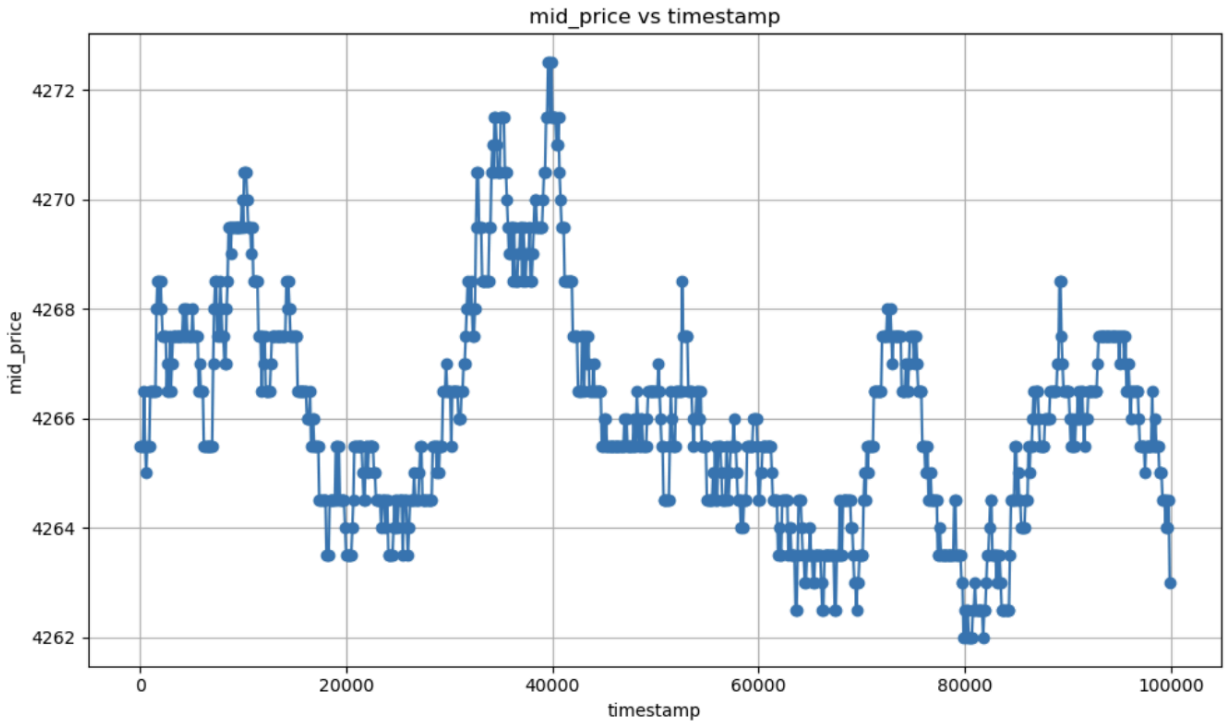


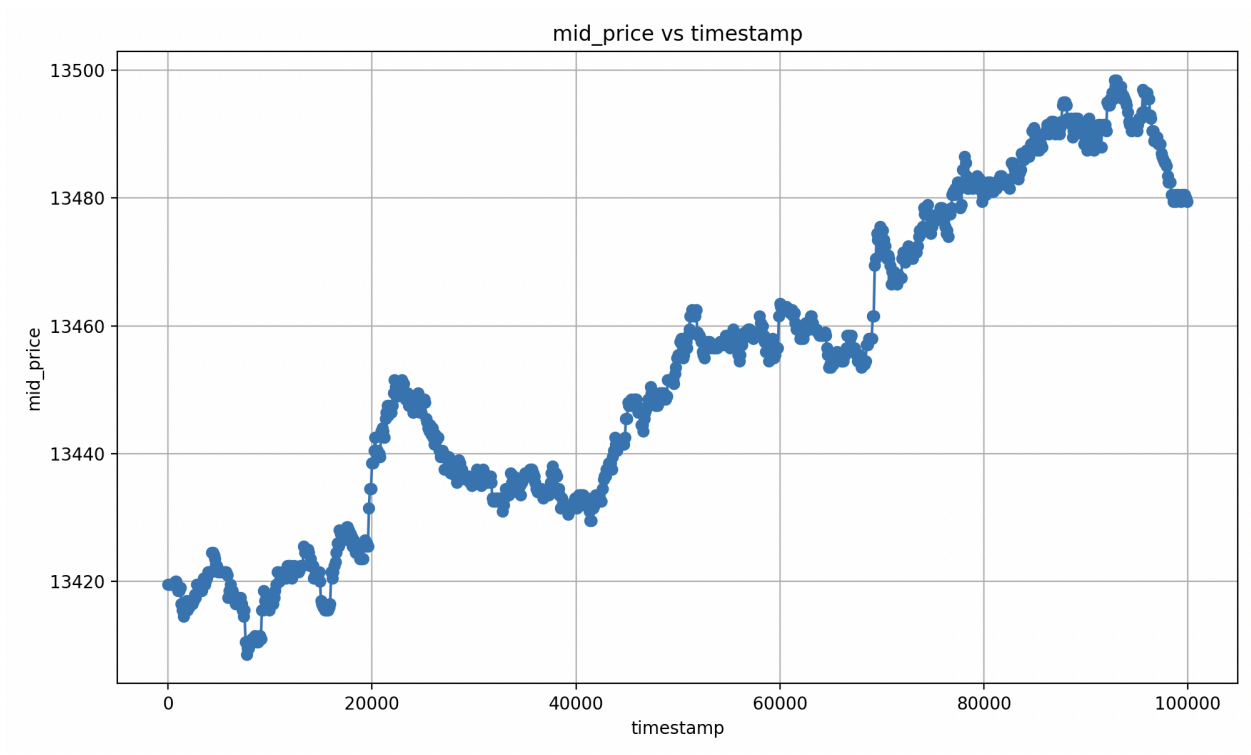
DATA ANALYSIS

PROSPERITY DATA (what we are tested on)

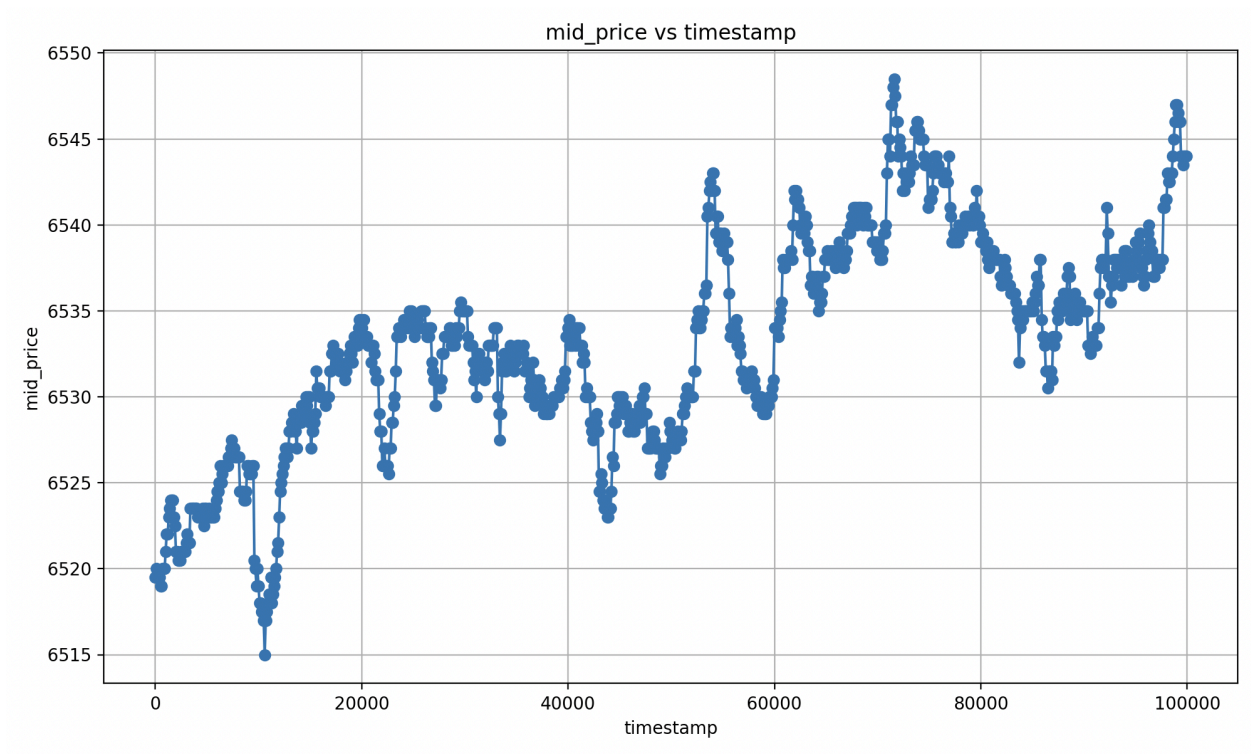
MID VS TIME croissants



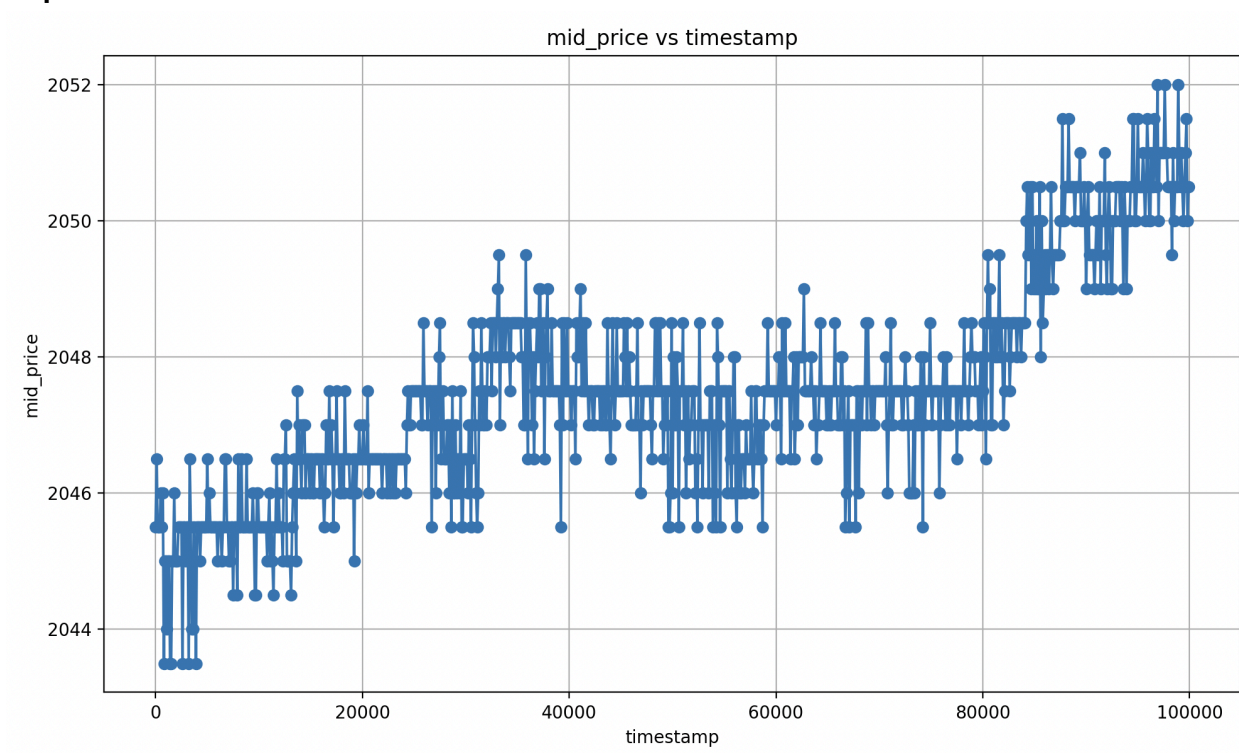
djembes



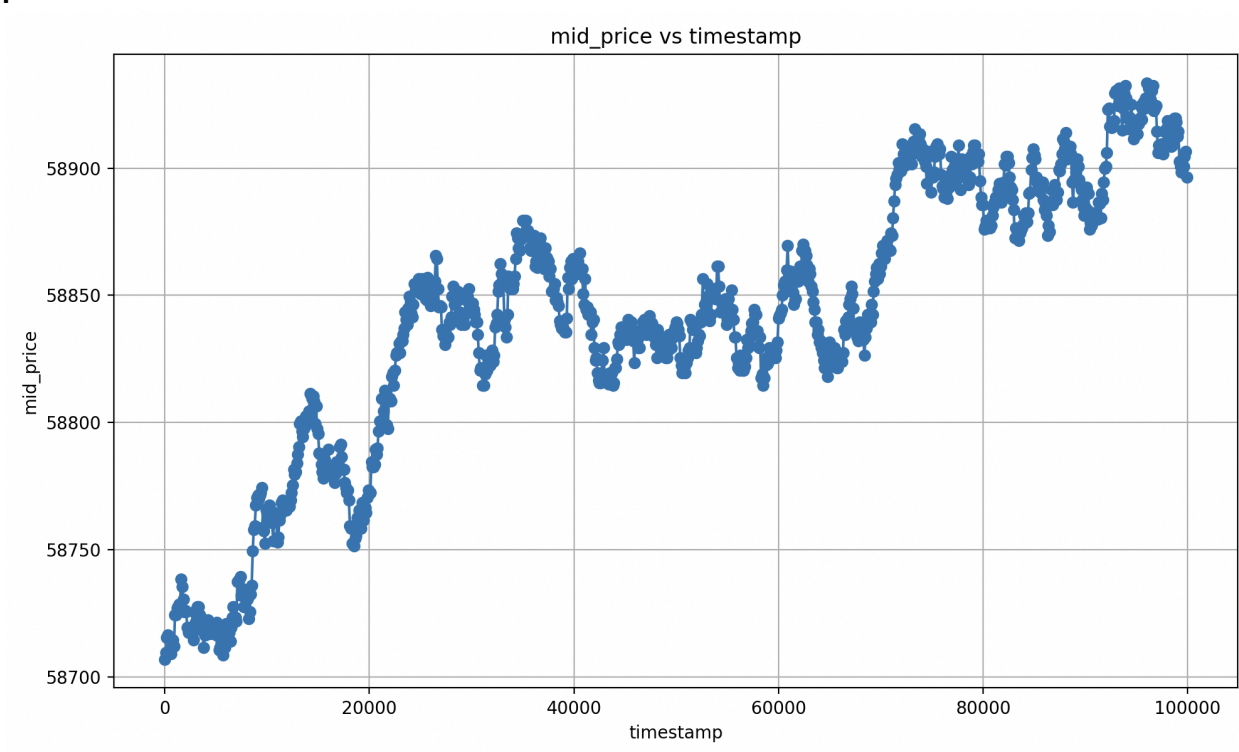
jams



kelp



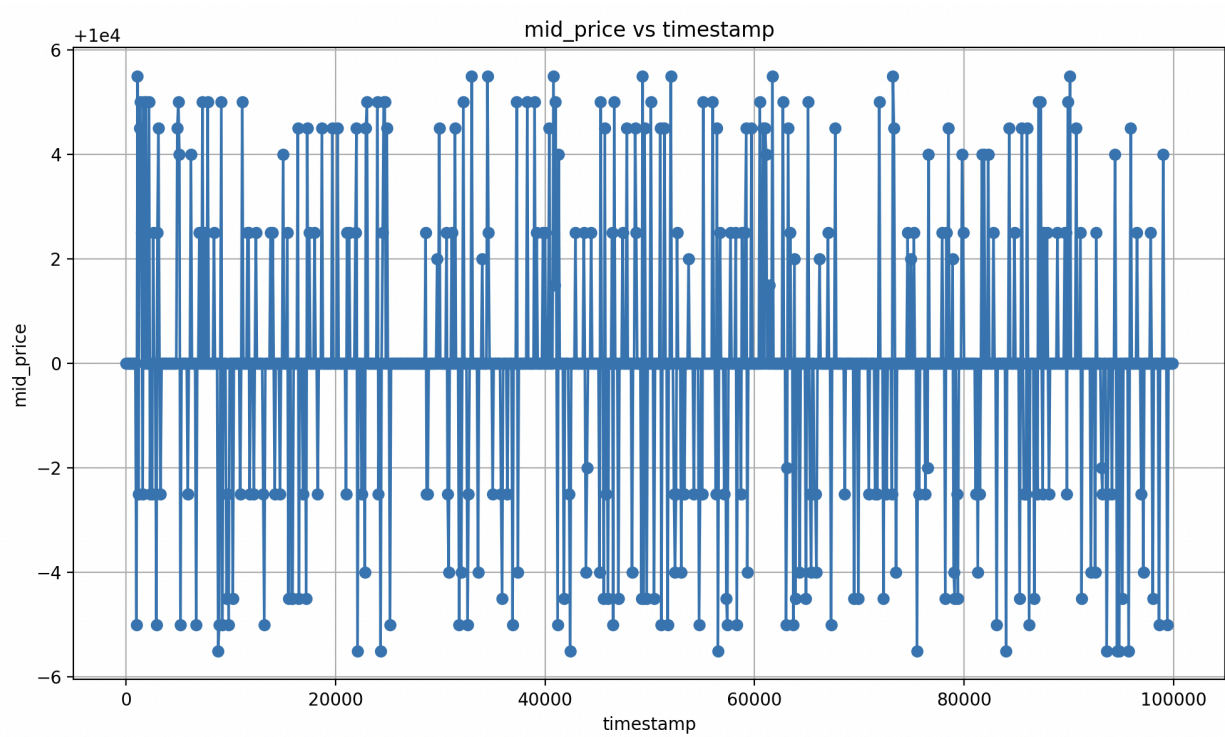
pb1



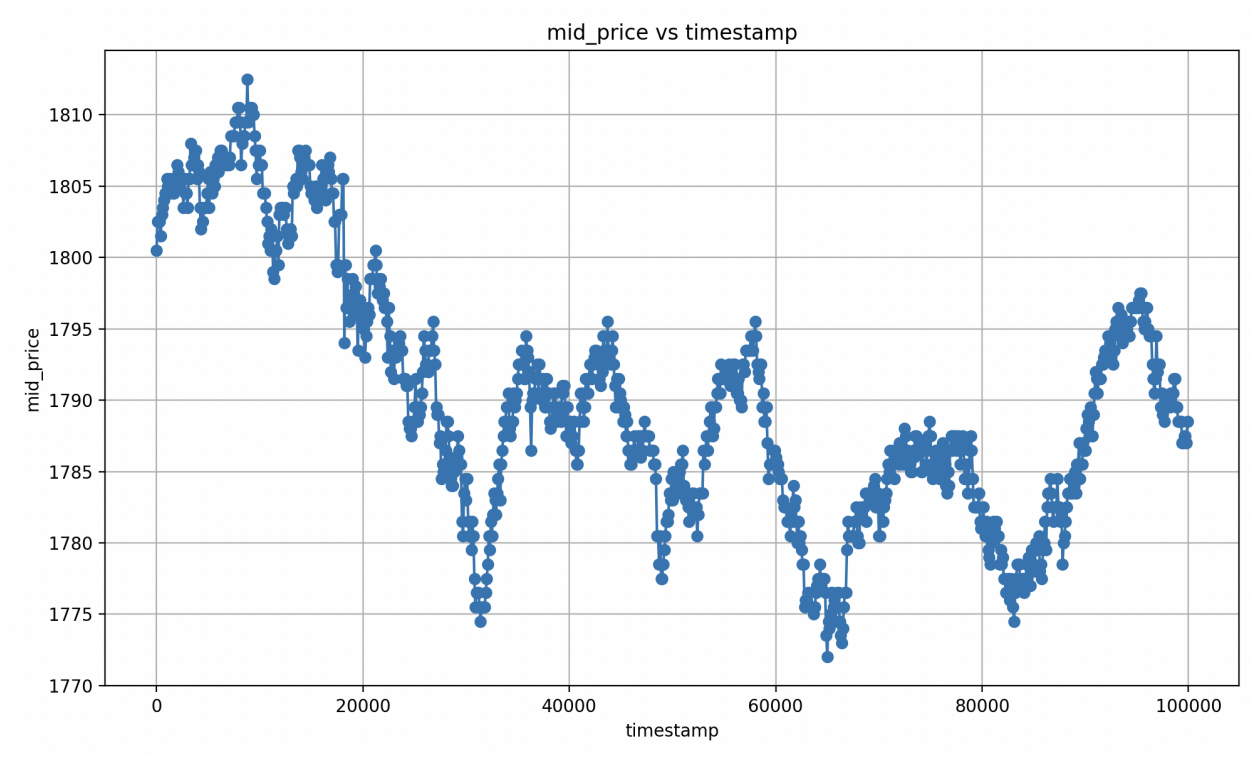
pb2



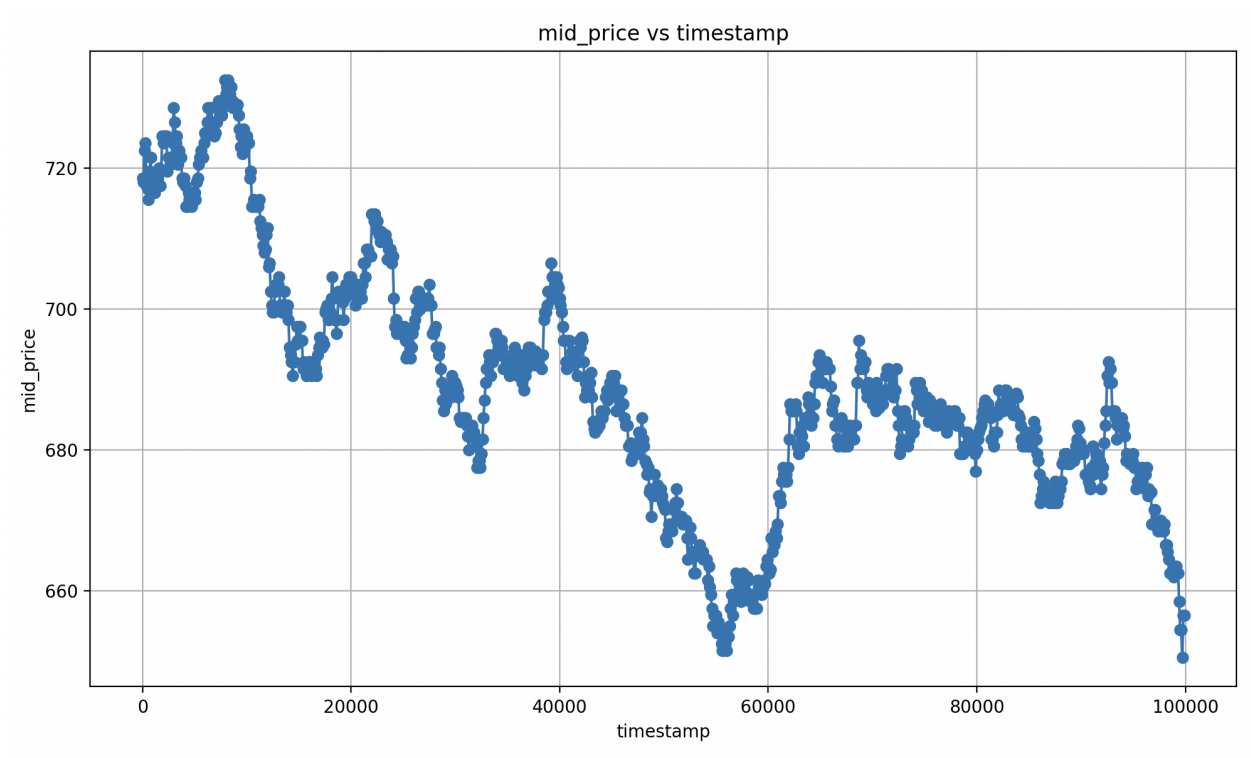
rr



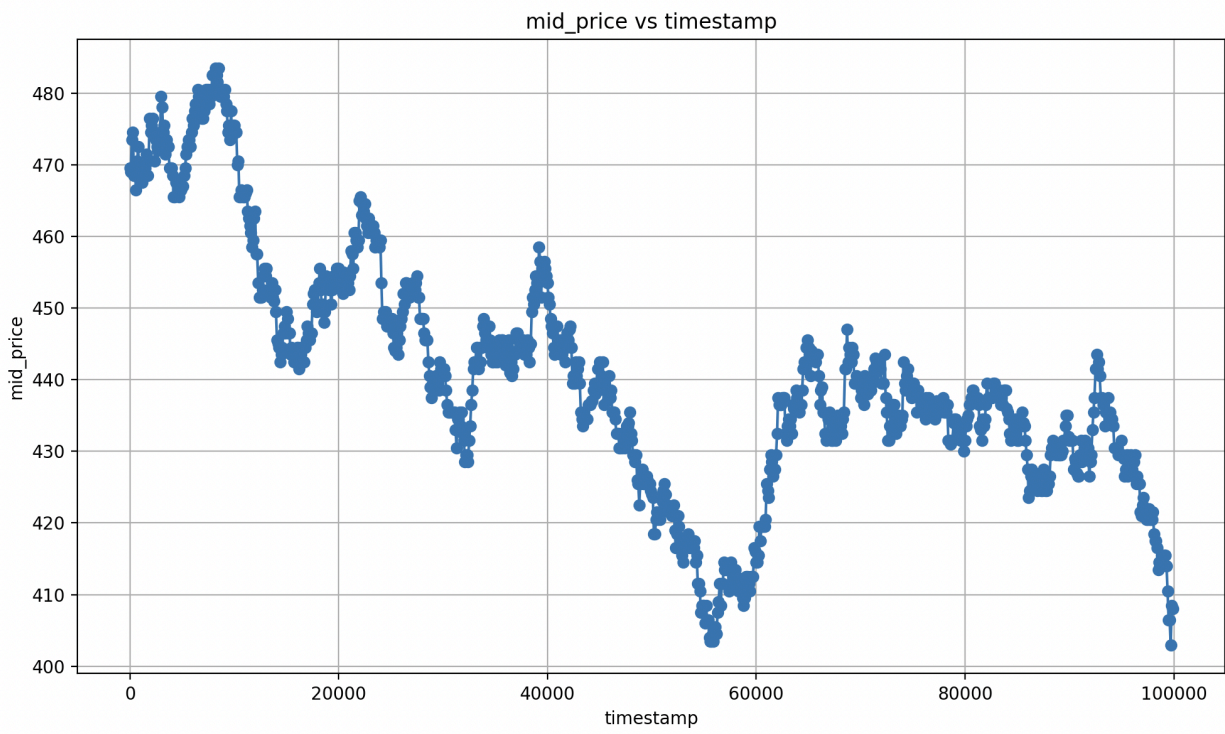
squink



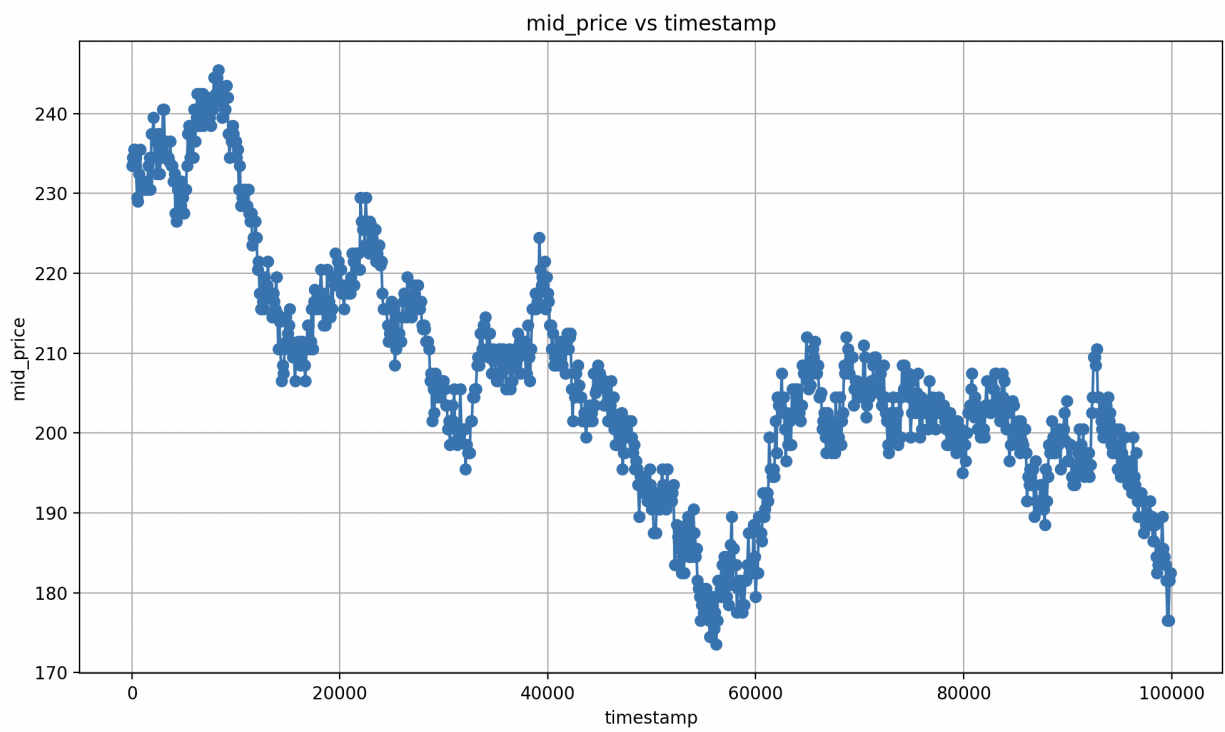
vr9500



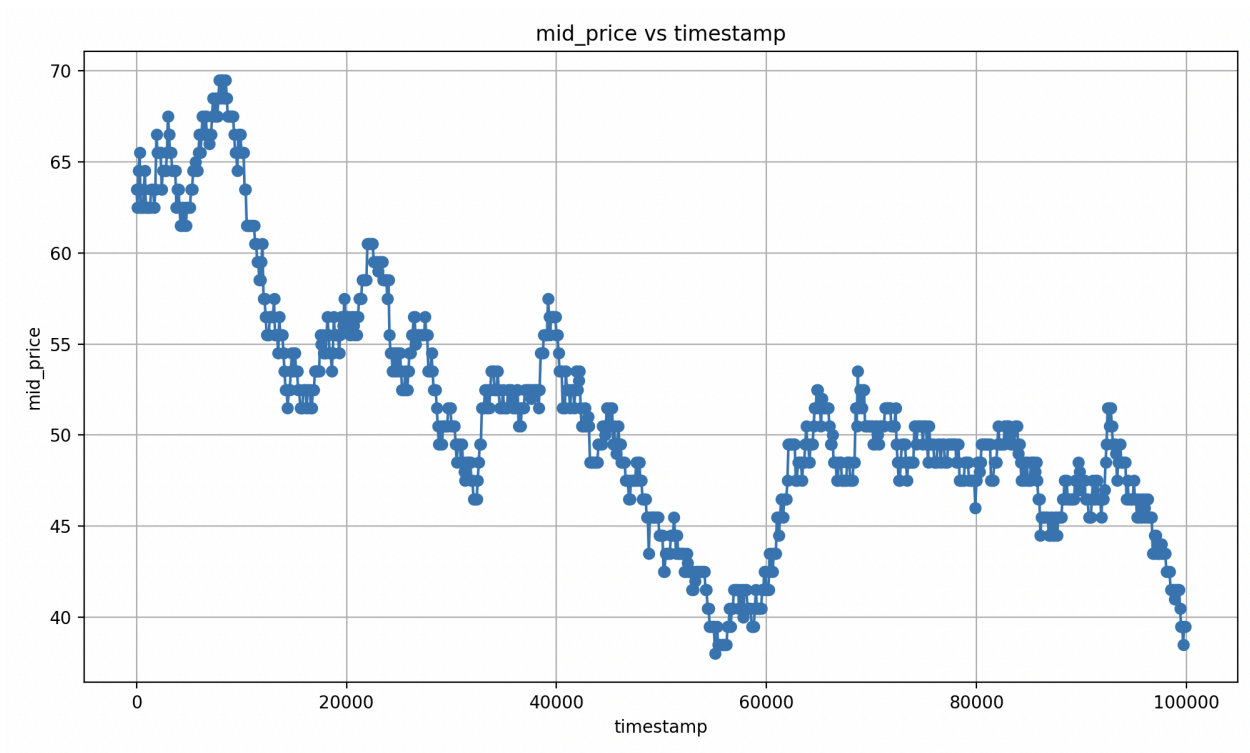
vr9750



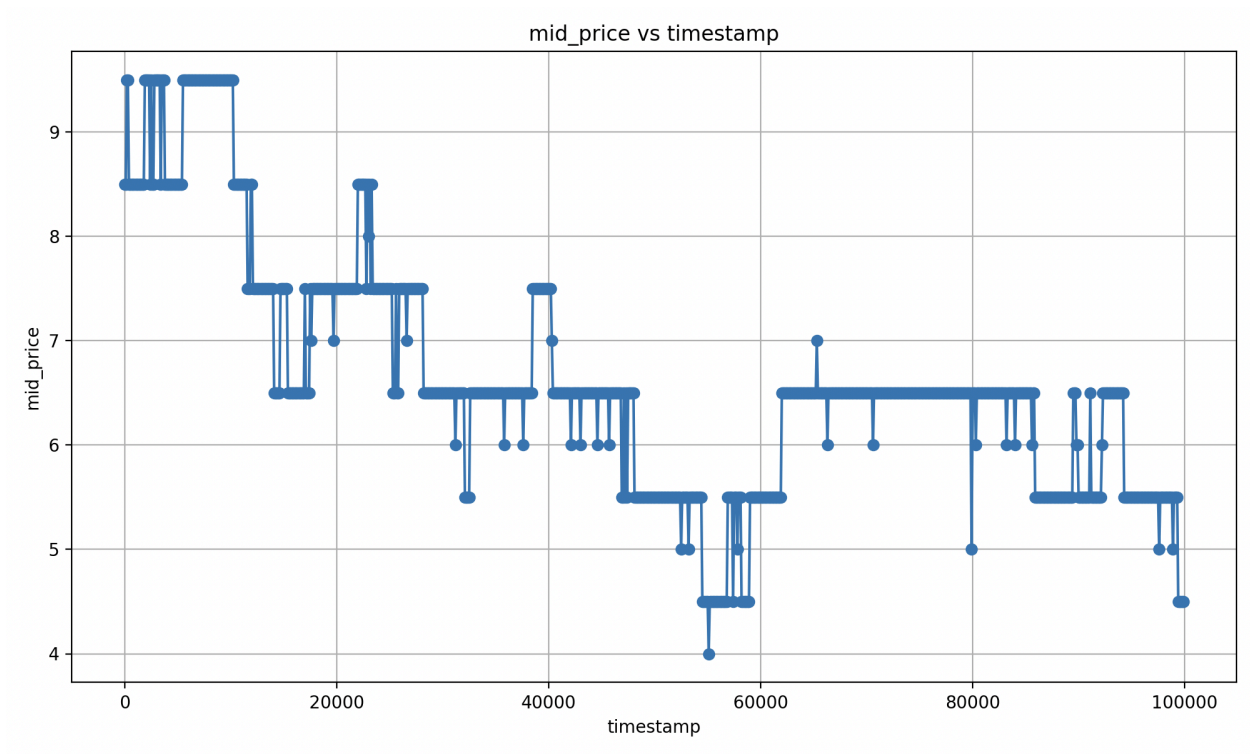
vr10000



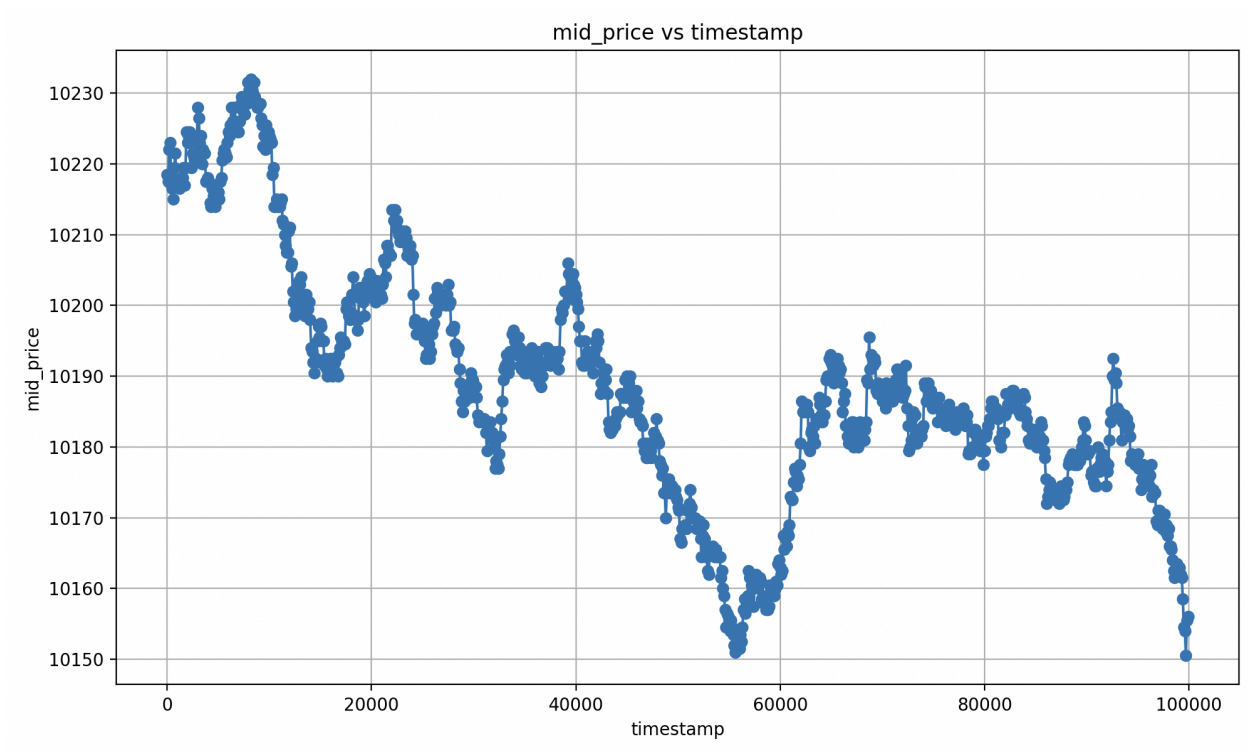
vr10250



vr10500



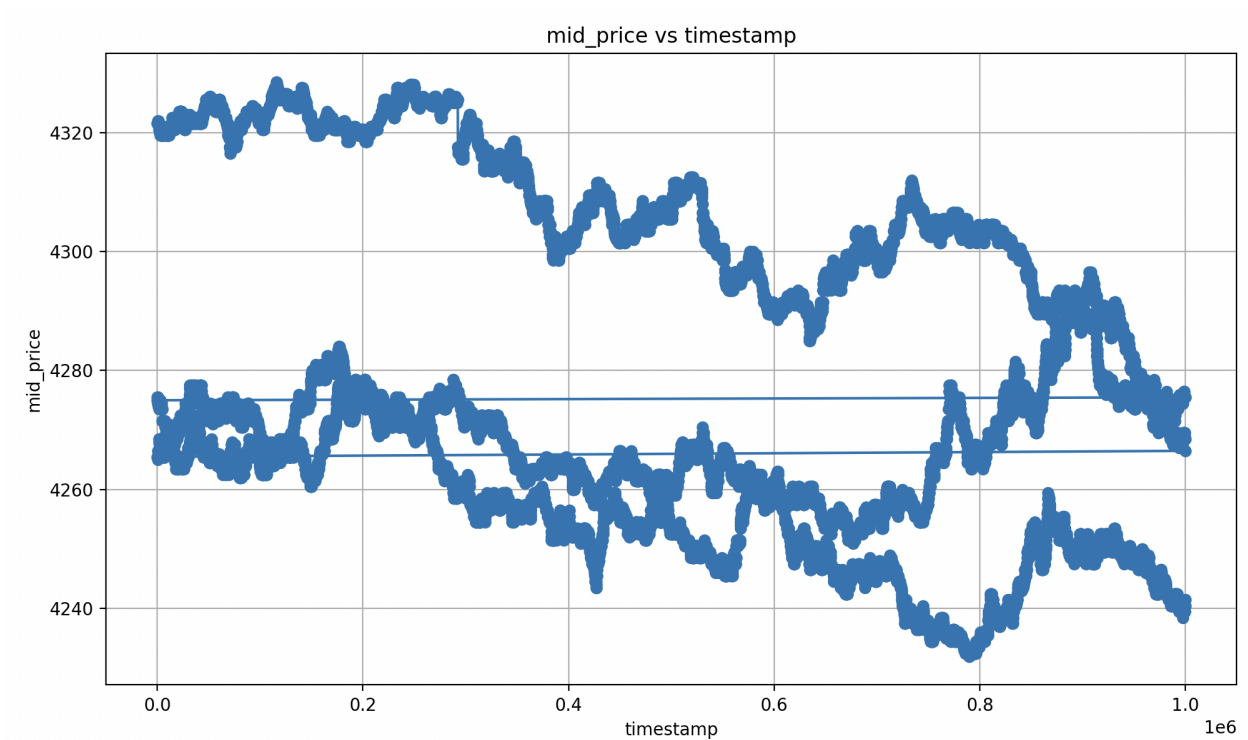
vr



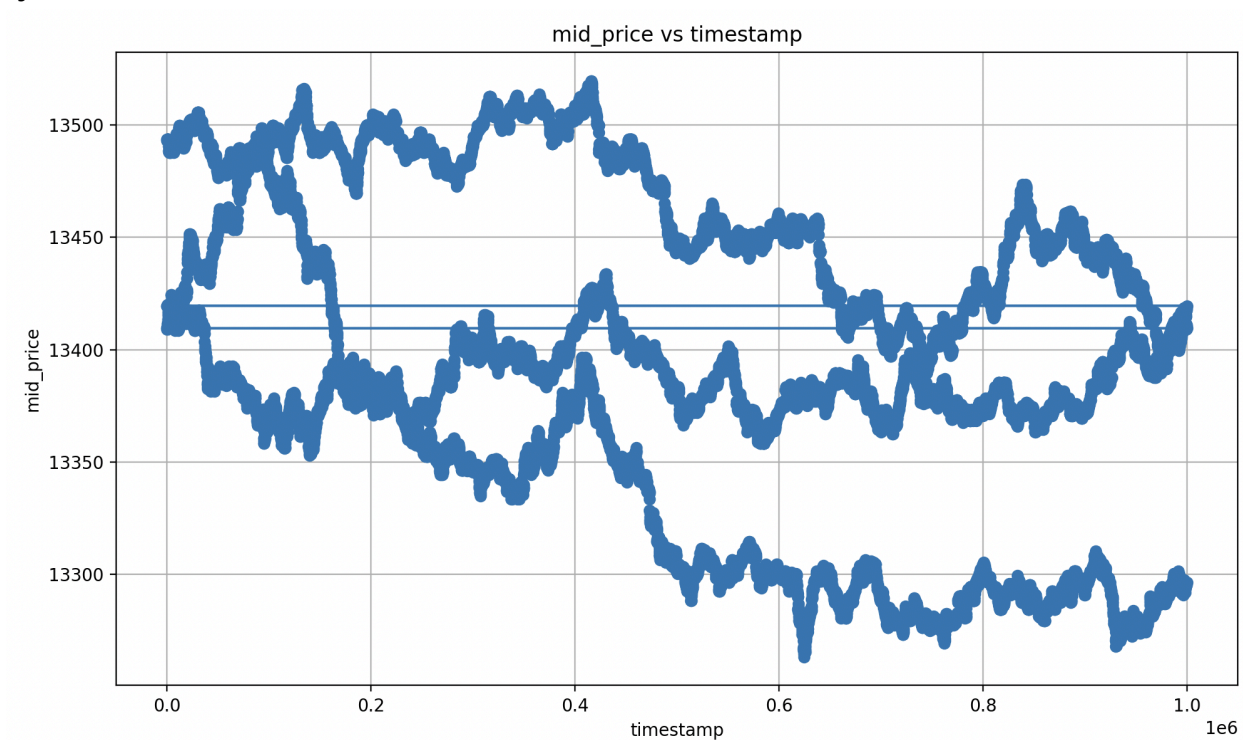
DATA IN A BOTTLE

(USING COMBINED DATA, SO ALL 3 DAYS ON CHART (DIF LINES))

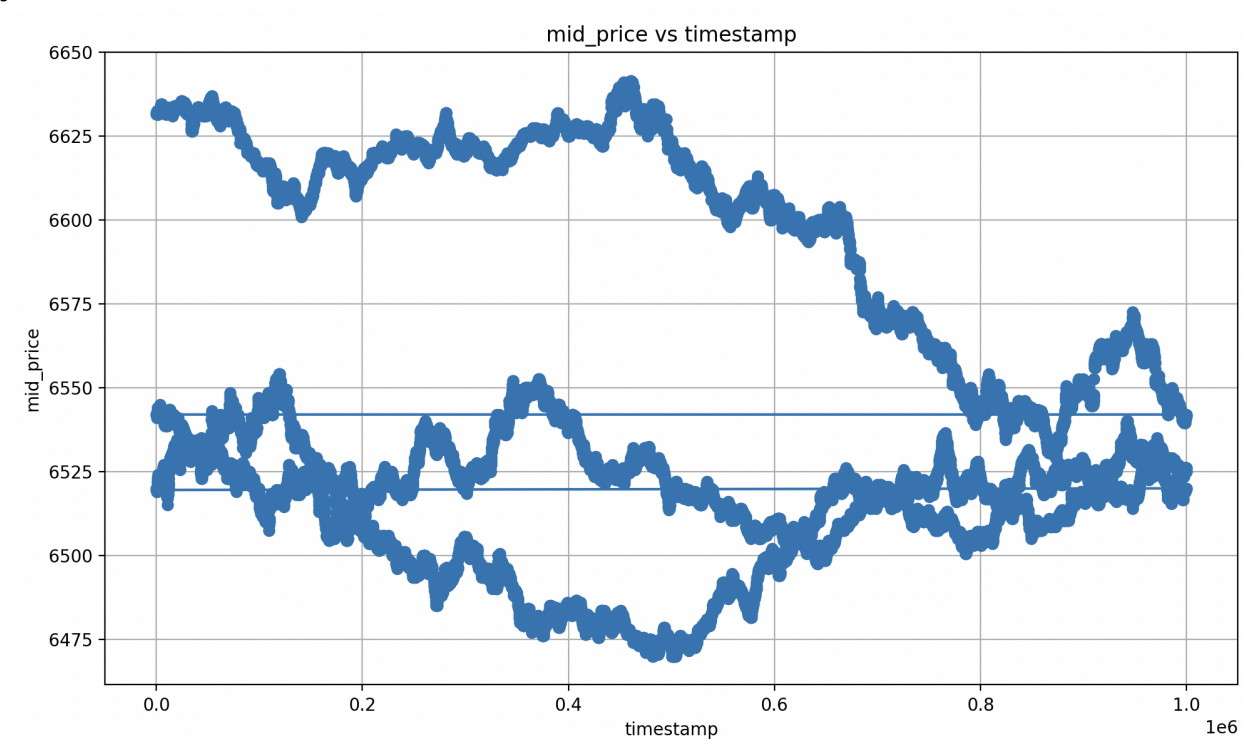
croissants



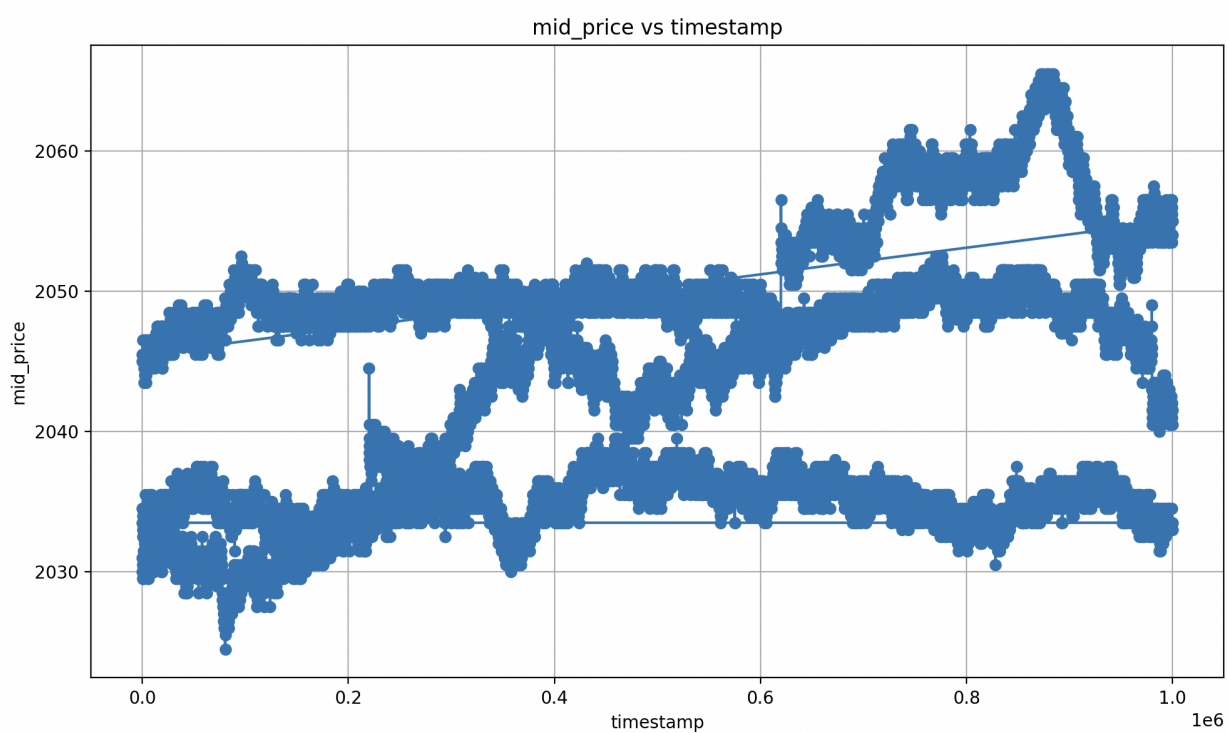
djembes



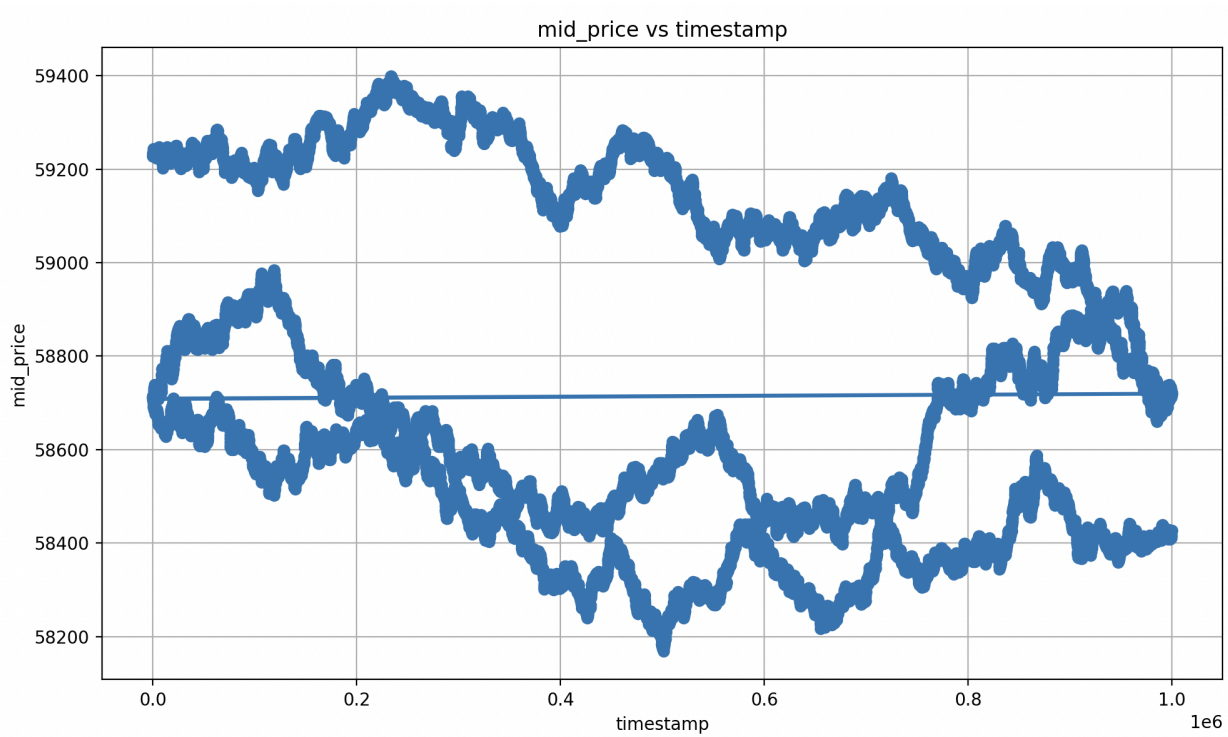
jams



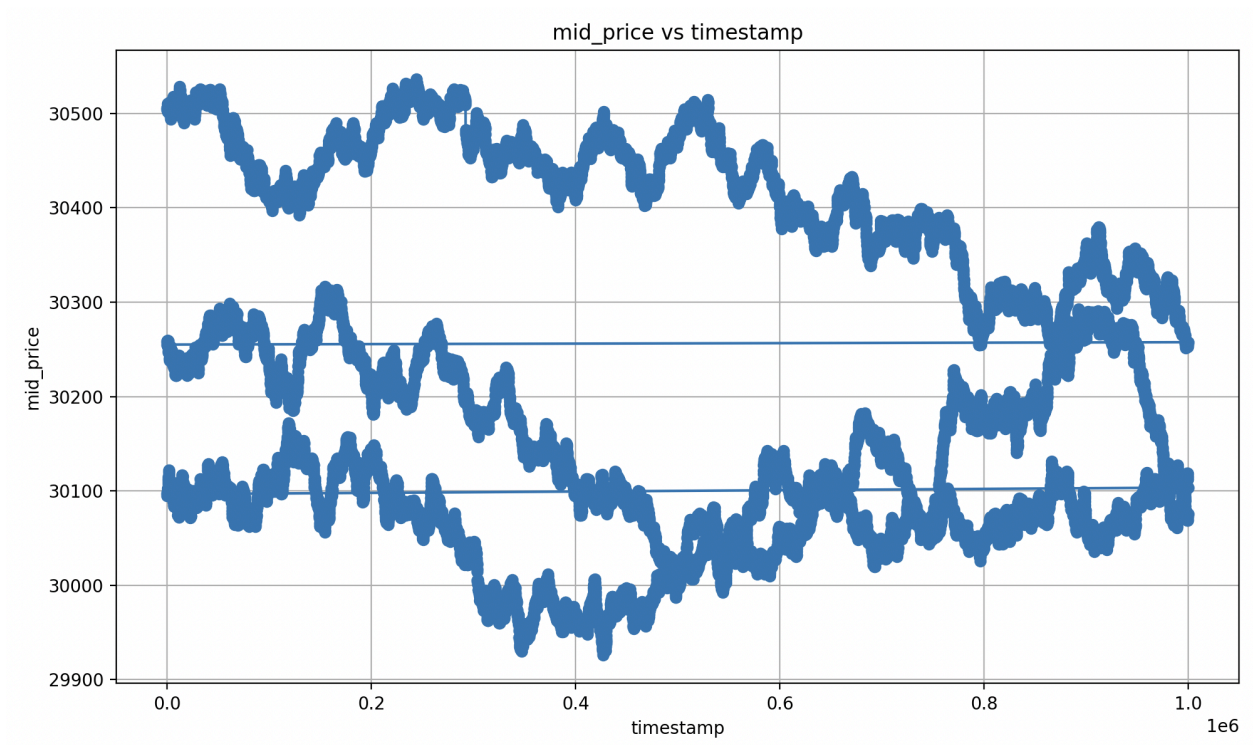
kelp



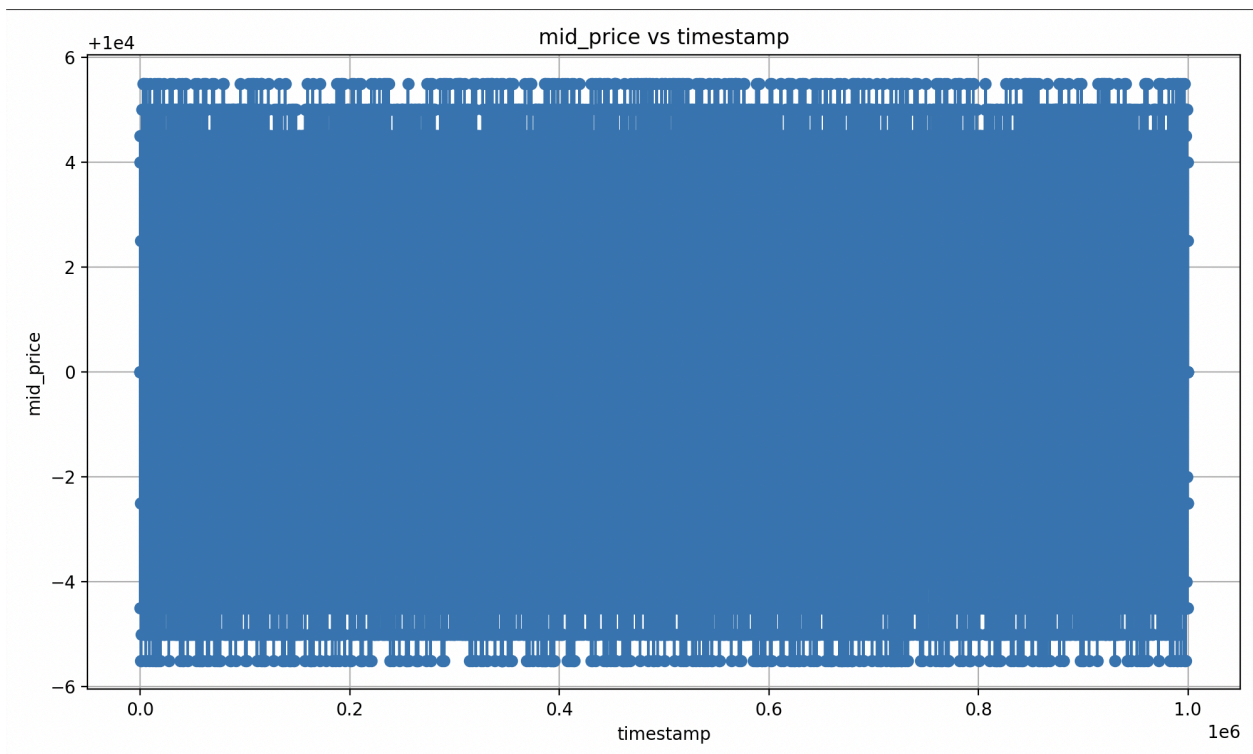
pb1



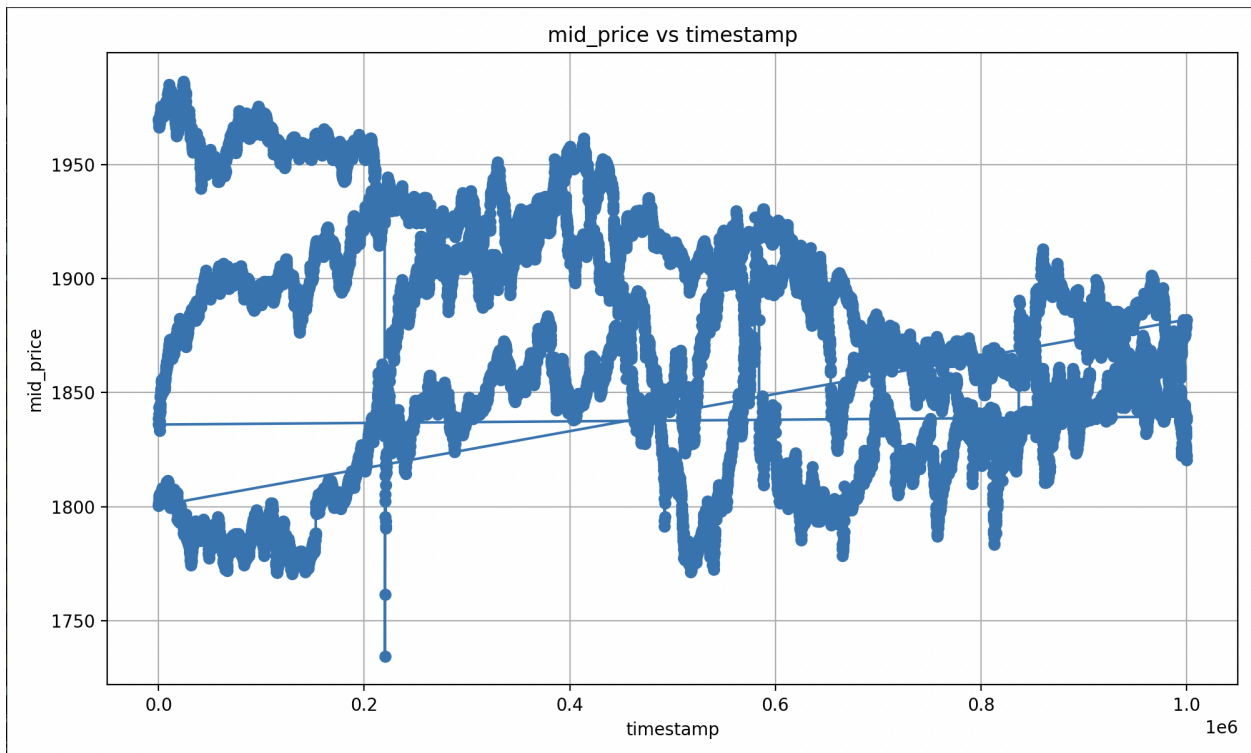
pb2



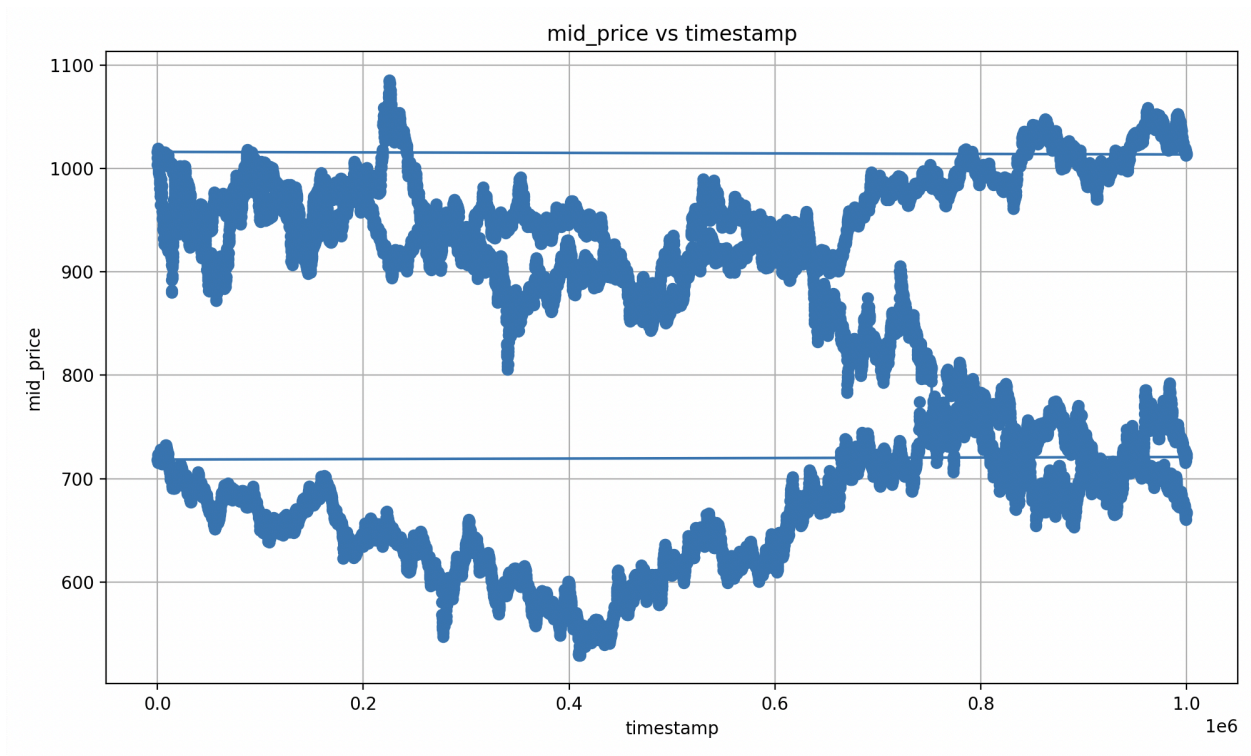
rr



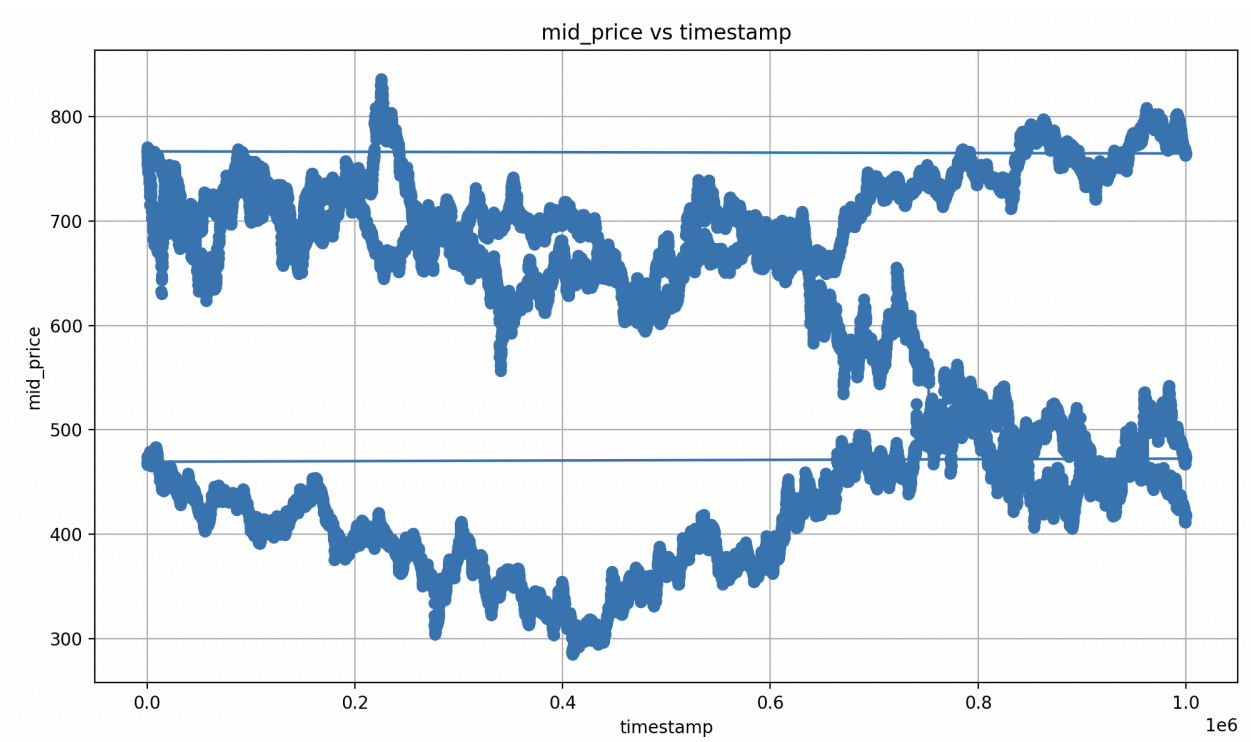
squink



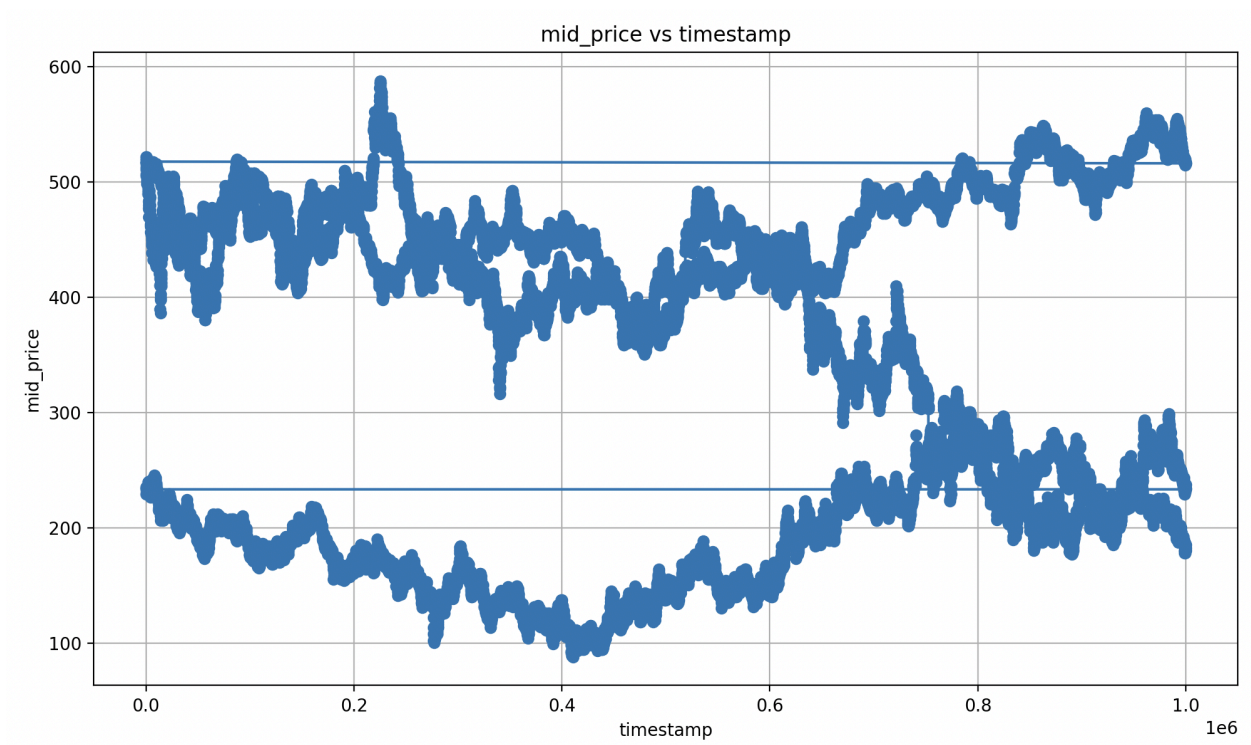
vr9500



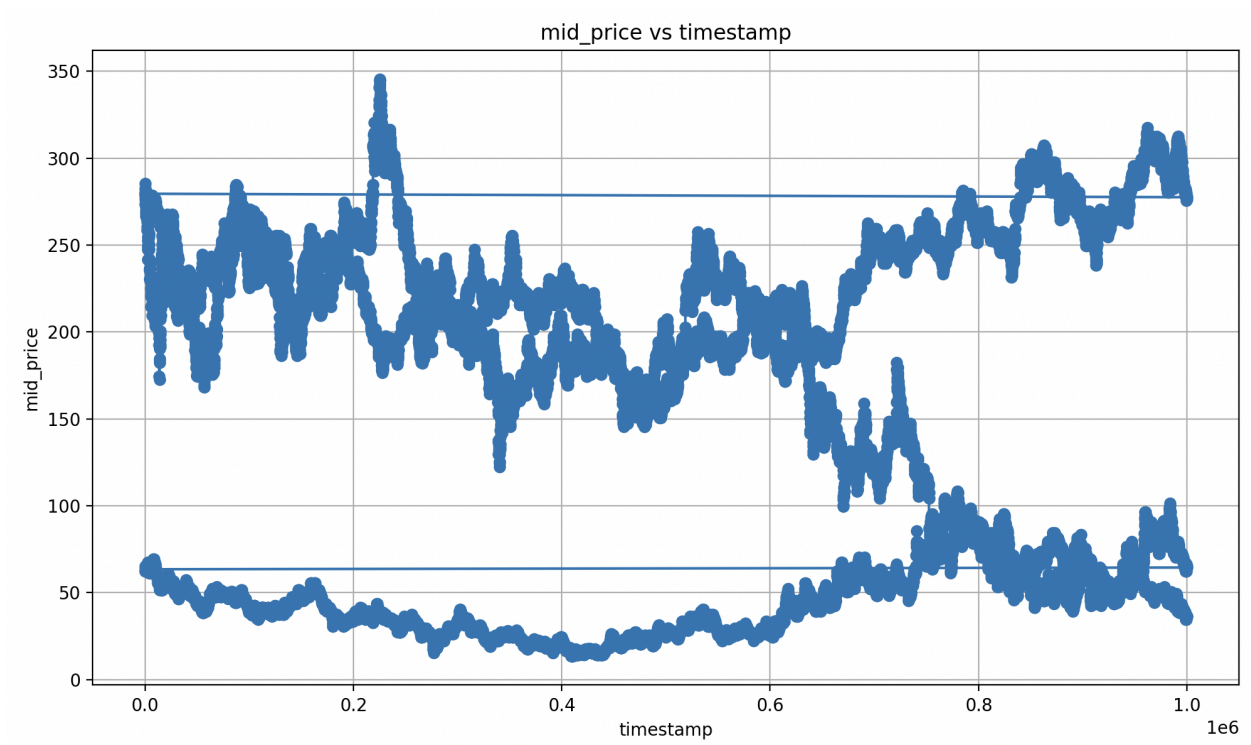
vr9750



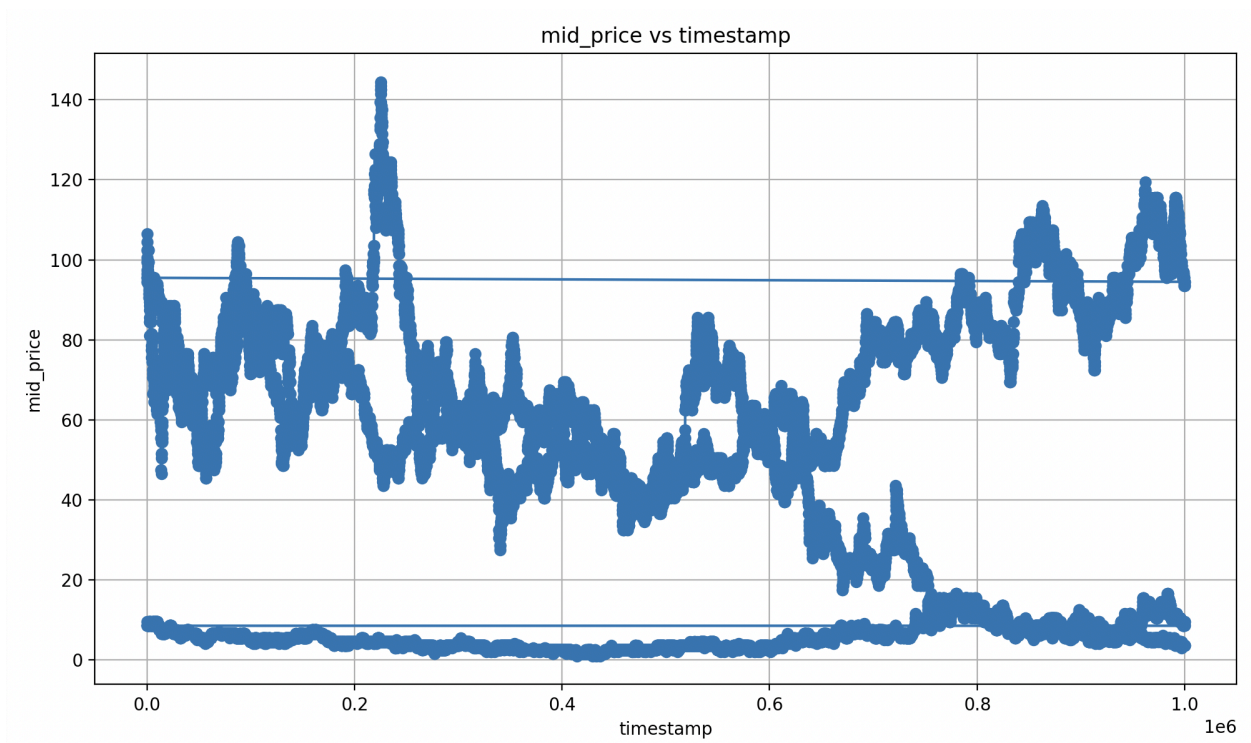
vr10000



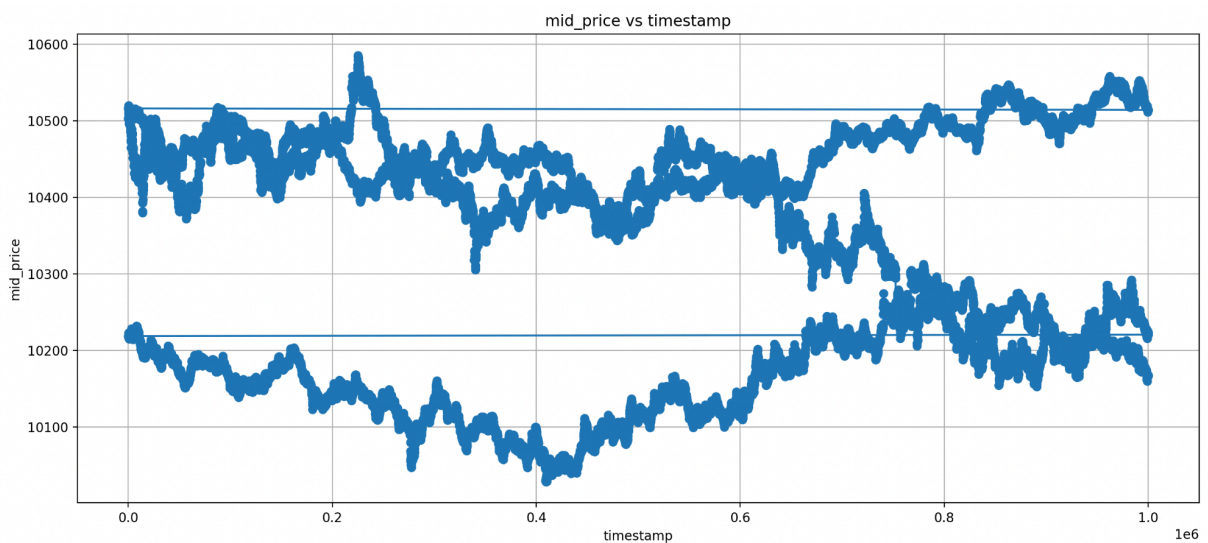
vr10250



vr10500



vr



CHAT GPT ANAL

🍌 Picnic Basket Components

CROISSANTS (CR)

- Trend: Price generally increases over time in all three days.
- Observation: Clear upward momentum, possibly driven by basket demand.
- Volatility: Moderate.
- Implication: Long positions may be favorable if momentum persists.

JAMS (JM)

- Trend: Prices show mild fluctuation with no strong directional movement.
- Observation: Range-bound behavior.
- Implication: Not ideal for momentum trading; better for basket arbitrage.

DJEMBES (DJ)

- Trend: Some fluctuations but less volatile.

- Observation: Appears relatively stable; slight dips then recoveries.
 - Implication: Good candidate for basket-to-individual arbitrage or synthetic price strategy.
-

Picnic Baskets

PICNIC_BASKET1 (PB1)

- Trend: Fluctuates but overall fairly stable.
- Observation: Moves similarly to the weighted average of CR, JM, and DJ.
- Implication: Basket arbitrage potential—watch deviations from its composite price.

PICNIC_BASKET2 (PB2)

- Trend: Slight uptrend with fewer dips than PB1.
 - Observation: Reflects the values of CR and JM; tracks CR spikes more strongly.
 - Implication: Correlate with CR heavily for arbitrage trades.
-

Core Round 1 Products

RAINFOREST_RESIN (RR)

- Trend: Very stable with minimal movement.
- Observation: Low risk, low reward.
- Implication: Suitable for conservative strategies.

KELP (K)

- Trend: Clear oscillating pattern, like a sine wave.
- Observation: High-frequency peaks and troughs.

- Implication: Ideal for short-term swing trades or mean-reversion strategy.

SQUID INK (SQINK)

- Trend: Exhibits cycles with intermediate up-and-down movement.
- Observation: Slightly more volatile than RR, less structured than K.
- Implication: Look for repeating patterns—statistical modeling opportunity.

Volcanic Rock Derivatives

VOLCANIC_ROCK (VR)

- Trend: Consistent rise across days.
- Observation: Strong bullish trend.
- Implication: Call vouchers increasing in value as expiry nears.

VOUCHERS (VR9500 → VR10500)

- Trend:
 - All show decay over time (as expected with options) but influenced by how far in the money they are.
 - Lower strike price vouchers (e.g., VR9500) maintain better value.
- Observation: The deeper in-the-money, the less steep the decay.
- Implication: Consider synthetic long strategies combining vouchers and spot VR.

Key Observations Across Assets

1. Volcanic Rock vouchers decay with time, but those that are in-the-money resist steep drops—this opens up calendar spreads or synthetic call strategies.
 2. CR and PB2 are highly correlated, while DJ and PB1 share more resemblance.
 3. Kelp's wave-like behavior is ideal for algo-trading using moving averages or momentum reversion.
 4. RR is extremely stable, functioning as a pseudo bond.
 5. VR is bullish, making call vouchers worth monitoring.
-

Next Steps in Data Analysis

1. Calculate Implied Basket Prices:
 - $PB1 \approx 6 \times CR + 3 \times JM + DJ$, $PB2 \approx 4 \times CR + 2 \times JM$.
 - Compare with market prices for arbitrage.
2. Estimate Volatility & Momentum:
 - Use rolling standard deviation to gauge volatility zones for each asset.
 - Apply SMA/EMA for K, SQINK, VR.
3. Option Payoff Modeling:
 - Simulate VR voucher payoffs by expiration (given strike prices).
 - Use this to back into theoretical value and find mispricings.
4. Statistical Correlation Matrix:
 - Run correlation coefficients between products and baskets.
 - Detect leading indicators.
5. Construct Synthetic Products:

- Combine individual CR/JM/DJ positions to replicate baskets.
- Combine vouchers + VR to simulate calls or straddles.